

GSM Architecture

Module 39

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Hello friends I welcome you all to the video lecture series of cyber security and information security in this lecture we will study about GSM system architecture first of all please see the contents which I shall cover after covering brief introduction about GSM I shall take up GSM system architecture in last I shall take up GSM services so now let us start global system for mobile communication that is gsm is a globally accepted standard for digital cellular communication digital seller networks are the segment of the market for mobile and wireless devices which are growing day to-day they are the wireless extensions of traditional pstn or ISDN networks and allow for seamless roaming with the same mobile phone Nation or even worldwide today these systems are mainly used for voice traffic however

data traffic is continuously growing and there are several Technologies for wireless data transmission using Cellular Systems the most popular digital system is gsm with approximately 70% market share GSM is the most successful digital mobile telecommunication system in the world it is used by over 800 million people in more than 190 countries in the early 1980s Europe had numerous coexisting analog mobile phone systems which were often based on similar standards but ran on slightly different carrier frequencies to avoid the situation for a second generation fully digital system the group special Mobile GSM was founded in 1982 this system was soon named the global system for mobile Communications

GSM with the specification process lying in the hands of European telecommunications Standard Institute etsi in the context of universal mobile telecommunication system UMTS and the creation of 3gpp third generation partnership project the whole development process of GSM was transferred to 3gpp and further development is combined with 3G development 3G PP assigned new numbers to all GSM standards the primary goal of GSM was to provide a mobile phone system that allows users to roam throughout Europe and provides voice Services compatible to ISDN and other pstn systems GSM is a typical second generation system replacing the first generation analog systems but not offering the high worldwide data rates that the third generation systems such as UMTS GSM has initially been deployed in

Europe using 890 to 915 MHz for uplinks and 935 to 960 MHz for down links now the system is also called GSM900 to distinguish it from the later versions these versions comprise GSM at 1800 MHz which is called DCS that is digital cellular system 1800 and the GSM system mainly used in us at 1900 MHz which is called PCS that is personal Communications service 9 1900 GSM system architecture GSM comes with a hierarchical complex system architecture comprising many entities and interfaces a GSM system consists of three subsystems radio subsystem RSS Network and switching subsystem NSS and operation subsystem that is OSS each subsystem will be discussed in more detail in the following sections generally a GSM customer only notices a

very small fraction of the whole network the mobile stations MS and some antenna of Base transceiver stations that is BTS Radio subsystem the radio subsystem RSS comprises all radio specific entities that is the mobile stations and the base station subsystem GSM architecture pair in the figure shows the connection between the RSS and the NSS via the A interface that is solid lines and the connection to the OSS via the O interface these are the dash lines the A interface is typically based on circuit switched PCM 30 systems carrying up to 3064 kilobits per second connections whereas the O interface uses the signaling system number seven based on X.25 carrying management data to and from the radio subsystem base station subsystem that is BSS a GSM network comprises many BSS

each controlled by a base station controller the BSS performs all functions necessary to maintain radio connections to an MS coding decoding of voice and rate adaptation to and from 6 minutes, 1 second the wireless network part the base station subsystem contains several base transceiver stations base transceiver station BTS a BTS comprises all radio equipment that is antennas signal processing amplifiers necessary for radio transmission a BTS can form a radio cell or using sectorized antennas and several cells are connected to Mobile station via the U interface and to the BSC while the Abis interface the U interface contains all the mechanisms necessary for wireless transmission that is TDMA FDMA ETC a GSM cell can measure between some 100 m and

35 kilometre depending on the environment base station controller that is BSC the BSC basically manages the BTS it reserves radio frequencies handles the Handover from one BTS to another within the BSS and performs paging of the MS the BSC also multiplexes the radio channels onto the fixed network connections at the A interface Mobile station that is MS mobile stations are most widely known cell or mobile phones are the section of a GSM cellular network that the user sees and operates in recent years their size has fallen dramatically while the level of functionality has greatly increased a further Advantage is that the time between charges has significantly increased there are number of elements to the cell phone although the two main elements are the main hardware and the SIM the hardware itself contains the

main elements of the mobile phone including the display case battery and the electronics used to generate the signal and process the data receiver and to be transmitted it also contains a number known as the international mobile equipment identity that is IMEI this is installed in the phone at the factory and cannot be changed it is accessed by the network during registration to check whether the equipment has been reported as stolen the SIM or subscriber identity module contains the information that provides the identity of the user to the network it contains a variety of information including a number known as the international mobile subscriber identity that is IMSI this SIM card contains many identifiers

and tables such as card type serial number a list of subscribed services a personal identity number PIN a PIN unblocking key P and authentication key Ki and the international mobile subscriber identity that is IMSI the PIN is used to unlock the MS using the wrong PIN three times will lock the SIM in such cases the P is needed to unlock the SIM the MS stores dynamic information while logged onto the GSM system such as example the cipher key KC and the location information consisting of a temporary mobile subscriber identity that is TMSI and location area identification that is LAI typical MS for GSM 900 have a transmit power of up to 2 W whereas for GSM 1800 1 W is enough due to the smaller size apart from the telephone interface an MS can also offer other types of

interfaces to users with display loudspeaker microphone and programmable soft keys further interfaces comprise computer modems or Bluetooth typical MS example mobile phones comprise many more vendor specific functions and components such as cameras fingerprint sensors calendars address books games and internet browsers personal digital assistance PDA with mobile phone functions are also available the listeners should be aware that an MS could also be integrated into a car or be used for location tracking of a container Network and switching subsystem the heart of the GSM system is formed by the network and switching subsystem that is NSS the NSS connects the wireless network with standard public networks and performs handovers between different base station subsystems it comprises

functions for worldwide localization of users and supports charging accounting and roaming of users between different providers in different countries the NSS consist of the following switches and databases mobile switching center that is MSC the main element within the core Network area of the overall GSM network architecture is the mobile switching center that is MSC the MSC acts like a normal switching node within a PSTN or ISDN but also provides additional functionality to enable the requirements of a mobile user to be supported they set up connections to other MSC's to and to BSC's via the A interface and form the fixed backbone network of a GSM system an MSC manages several BSC's in a geographical region

these include registration authentication call location inter MSC handovers and call routing to a mobile subscriber it also provides an interface to PSTN so that calls can be routed from the mobile network to a phone connected to a landline interfaces to other MS's are provided to enable calls to be made to mobiles on different networks MSC can also connect to public data

networks PDN such as x.25 an MC handles all signaling needed for connection setup connection release and Handover of connections to other ms's home location register hlr the hlr is the most important database in a GSM system it stores all user relevant information this comprises static information such as the mobile subscriber ISDN number that is msisdn subscribe Services example call

forwarding roaming restrictions GPRS and the international mobile subscriber identity that is imsi damic information is also needed example the current location area of the MS the mobile subscriber roaming number that is msrn the current vlr and MSC as soon as an MS leaves its current LA the information in the hlr is updated when a user switches on their phone the phone registers with the network and from this it is possible to determine which BTS it communicates with so that incoming calls can be routed appropriately even when the phone is not active but switched on it re-registers periodically to ensure that the network hlr is aware of its latest position this information is necessary to localize a user in the world while G M Network all these user specific information elements only exist once for

each user in a single hlr which also supports charging and accounting hlrs can manage data for several million customers and contain highly specialized databases which must fulfill certain realtime requirements to answer requests within certain time bounds there is one HL per Network although it may be distributed across various subcenters to for operational reasons visitor location register that is VR the vlr associated to each MSC is the dynamic database which stores all information needed for the MS users currently in the location area that is associated to the MSC example I MSI msisdn hlr address if a new Mobile station comes into a location area La the VR is responsible for this process and it copies all

relevant information for this user from home location register this hierarchy of vlr and hlr avoids frequent hlr updates and long distance signaling of user information the vlr can be implemented as a separate entity but it is commonly realized as a integral part of the msse rather than a separate entity in this way exess is made faster and more convenient equipment identity register e the E is the entity that decides whether given mobile equipment may be allowed onto the network each mobile equipment has a number known as the international mobile equipment identity this number is installed in the equipment and is checked by the network during registration authentication Center Au the Au is a protected database that contains the secret key also contained in the users's SIM card it is used for

authentication and for ciphering on the radio channel Gateway mobile switching Center gmsc the GM SC is the point to which a mobile equipment terminating call is initially routed without any knowledge of the MS location the gmsc is thus in charge of obtaining the msrn that is Mobile station roaming number from the hlr based on the msisdn Mobile station ISDN number the directory number of a Mobile station and routing the call to correct visited mobile switching service center the MSC part of the term gmsc is misleading since the Gateway operation does not require any linking to an MSC SMS gateway msg the msg or SMS

gateway is the term that is used to collectively describe the two short message Services gateways defined in the GSM standards the two gateways handle

messages directed in different directions the SMS gsmc short message service Gateway mobile switching Center is for short messages being sent to a mobile equipment the SMS iwmc short message service interworking mobile switching Center is used for short messages originated with the mobile on that Network the SMS gsmc role is similar to that of gsmc whereas the SMS iwmc provides a fixed access point to the short message service center operation subsystem the third part of a GSM system the operational subsystem that is OSS contains the necessary functions for Network operation and maintenance the old SS processes its own network entities and accesses other entities via ss7 signaling the operation support subsystem is an element within the

overall gsm network architecture that is connected to components of the NSS and the BSC it is used to control and monitor the overall gsm network and it is also used to control the traffic load of the BSS it must be noted that as the number of BS increases with the scaling of the subscriber population some of the maintenance tasks are transferred to the BTS which allowing Savings in the cost of ownership of the system the following entities have been defined operation and maintenance center that is OMC the OMC monitors and controls all other network entities via the O interface typical OMC management functions are traffic monitoring status 20 minutes, 1 second reports of network entities subscriber and Security Management or accounting

and billing OMC has use the concept of telecommunication management Network that is tmn as standardized by the

itu authentication center that is au as a radio interface and mobile stations are particularly vulnerable a separate Au has been defined defined to protect user identity and data transmission the Au contains the algorithms for authentication as well as the keys for encryption and generates the values needed for user authentication in the hlr the Au May in fact be situated in a special protected part of the hlr equipment identity register that is eir the eir is a database for all IM EIs that is it stores all device identifications registered for this network as Ms are mobile they can be easily stolen with the valid Sim anyone could use the stolen Ms the E has a blacklist of stolen or locked devices Unfortunately The Blacklist of different providers are not usually

synchronized and the illegal use of a device in another operator network is possible the equipment identity register also contains a list of valid imis white list and a list of malfunctioning devices that is gy list GSM Services GSM offers much more than just voice telephony contact your local gsm network operator to the specific services that you can Avail GSM offers the three basic types of services telephone Serv services or T services data services or beer Services supplementary Services T services the abilities of a bearer service are used

by a T service to transport data these services are further transited in the following ways voice calls the most basic T service supported by GSM is telephony this includes full rate speech at 13 kbps and emergency calls where the

nearest emergency service provider is notified by dialing three digits video text and Facsimile another group of T services include video text access to text transmission Facsimile alternate speech and automatic Facsimile smiling group short text messages short messaging services SMS service is a text messaging service that allows sending and receiving text messages message on your GSM mobile phone in addition to simple text messages other text Data including News sports Financial language and location based data can also be transmitted Bearer Services data services or Bearer services are used through a GSM phone to receive and send data is the essential building block leading to widespread mobile internet access and mobile data transfer

GSM currently has a data transfer rate of 9.6 kbps new development that will push up the data transfer rates for GSM users are HSCSD that is highspeed circuit switch data and GPRS that is General packet Radio Service are now available supplementary Services supplementary services are additional services that are provided in addition to T services and Bearer 24 minutes, 1 second Services these Services include caller identification call forwarding call waiting multi-party conversations and barring of international outgoing calls among others a brief description of supplementary Services is given here conferencing it allows a mobile subscriber to establish a multi-party conversation that is a simultaneous

conversation between three or more subscriber to set up a conference call this service is only applicable to normal telephoning call waiting this service notifies a mobile subscriber of an incoming call during a conversation the subscriber can answer reject or ignore the incoming call call hold this service allows a subscriber to put an incoming call on hold and resume after a while the call hold service is applicable to normal telephony call forwarding call forwarding is used to divert calls from the original recipient to another number it is normally set up by the subscriber himself it can be used by the subscriber to divert calls from Mobile station when the subscriber is not available call

barring call barring is useful to restrict certain types of outgoing calls such as ISD or stop incoming calls from undesired numbers call barring is a flexible service that enables the subscriber to conditionally bar calls number identification there are following supplementary Services related to number identification call line identification presentation this service displays the telephone number of the calling party on your screen calling line identification restriction a person not wishing their number to be presented to others subscribes to this service connected line identification presentation this service is provided to give the calling party the telephone number of the person to whom they are

connected this service is useful in situations such as forwardings where the number connected is not number dialed connected line identification restriction there are times when the person called does not wish to have their number presented and so they would subscribe to this person normally this overrides the presentation service malicious call identification the malicious call identification service was provided to combat thead of obscene or annoying calls the victim should subscribe to this service and then they could cause unnown malicious calls to be identified in the gsm network using a simple command advice of charge AOC this service was designed to give the subscriber an indication of the cost of the services that they are used

furthermore those service providers who wish to offer rental services to subscribers without their own Sim can also utilize this service in a slightly different form AOC for data calls is provided on the basis of time measurements closed user groups cugs this service is meant for groups of subscribers who wish to call only each other and no one else so friends this was all about GSM system architecture hope the concepts explained in this lecture were understandable and helpful hope to see you in the next lecture till then goodbye thank you